

Aditya Saindane

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AI/ML Engineer | Computer Vision | Full-Stack Development

PROFESSIONAL SUMMARY

AI/ML-focused Computer Science undergraduate with hands-on experience in computer vision, deep learning, backend development, and full-stack web applications. Built U-Net based image deraining, SAR image colorization, traditional image recognition, and multimodal LLM-guided 3D geometry workflows. Currently exploring local-first agentic desktop systems that combine automation, multimodal AI, secure application layers, and Windows-based productivity workflows.

EDUCATION

- D.Y. Patil International University** Pune, India
Bachelor of Technology - Computer Science; Current CGPA: 7 September 2022 - Present
- Rustomjee International School** Jalgaon, India
Class 12th (PCM), CBSE board; Percentage: 81 2021 - 2022
- Rustomjee International School** Jalgaon, India
Class 10th, CBSE board; Percentage: 88.6 2019 - 2020

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript, HTML, CSS
- AI/ML and Computer Vision:** PyTorch, TensorFlow, OpenCV, CNNs, U-Net, Image Restoration, Image Processing, RAG, Multimodal LLMs
- Libraries:** NumPy, Scikit-learn, Scikit-image, Matplotlib
- Web and Backend:** React, Node.js, Django, REST APIs, MongoDB
- Databases and Tools:** MongoDB, MySQL, SQLite, Git, GitHub, VS Code, Jupyter Notebook, MATLAB, Arduino IDE
- Core Strengths:** Critical Thinking, Effective Communication, Collaboration, Resilience, Conflict Resolution

EXPERIENCE

- Shobi Group** Pune
Backend Developer June 2024 - July 2024
 - Developed and deployed Django-based web applications using Python.
 - Designed database components and backend logic for application workflows and front-end integration.
- MANIT** Bhopal
Machine Learning Engineer May 2025 - July 2025
 - Developed a deep learning-based image deraining model using a modified U-Net architecture.
 - Applied hybrid filtering and evaluated output using PSNR, MSE, and SSIM image-restoration metrics.

PROJECTS

- Multimodal LLM-Guided 3D Geometry Analyzer (Completed)**
Tech: Python, Panda3D, PyTorch, Multimodal LLMs, Vision Models, RAG, NumPy
 - Architected a 3D model generation framework that combines custom geometry representations with vision-based ML systems for structured model analysis and generation.
 - Designed a vertex-edge-face format for object shape and spatial constraints; rendered outputs in Panda3D and integrated RAG with multimodal LLMs for semantic reference extraction.
 - Planned modular pipelines for model ingestion, geometry validation, visual rendering, semantic tagging, and prompt-controlled scene refinement.
- Local Agentic Desktop Workspace Layer (Completed)**
Tech: Python, Windows APIs, Local LLMs, RAG, Automation Agents, React, FastAPI, SQLite
 - Conceptualized a Windows-based productivity layer that behaves like an AI-assisted workspace without replacing the operating system or compromising core system security.
 - Designed a local-first architecture with a desktop shell interface, sandboxed task agents, file-aware retrieval, command approval gates, and isolated automation modules.
 - Planned components for multimodal input, contextual memory, project-level file indexing, code execution support, app-launch workflows, and secure human-in-the-loop actions.
 - Defined a phased implementation roadmap covering prototype UI, local knowledge base, agent router, task planner, safety layer, plugin APIs, and evaluation workflows.
- Deep Learning Deraining System (Completed)**
Tech: Python, PyTorch, OpenCV, NumPy, Scikit-image

- Built a single-image deraining system using a modified U-Net encoder-decoder architecture with skip connections to remove rain streaks while preserving scene structure.
- Trained on paired rainy-clean datasets with MSE, perceptual, and adversarial losses; applied curvature-aware post-processing and evaluated with PSNR, MSE, and SSIM.
- Implemented preprocessing, augmentation, inference visualization, and metric-reporting utilities to compare rainy inputs, restored outputs, and clean references.
- **Blinking Cosmos | Full-Stack Web Application**
Tech: HTML, CSS, JavaScript, Node.js, MongoDB, React
 - Spearheaded development of an interactive astronomy platform covering celestial bodies, astronomical phenomena, universe facts, and cosmic theories.
 - Built responsive React interfaces and connected Node.js/MongoDB backend components to organize content, support dynamic pages, and improve cross-device usability.
 - Structured reusable front-end sections and backend routes to simplify content updates and improve navigation across the application.
- **SAR Image Colorization Model (Beta)**
Tech: Python, TensorFlow, OpenCV, Scikit-learn
 - Engineered a machine learning pipeline to generate colorized Synthetic Aperture Radar images from grayscale SAR inputs using image-processing workflows.
 - Implemented preprocessing, enhancement, and training workflows to improve visual clarity, reduce artifacts, and support easier SAR interpretation.
 - Explored filtering, contrast normalization, and feature-preserving transformations to strengthen visual output quality during experimentation.
- **Mathematical Image Recognition Model**
Tech: Python, OpenCV, NumPy, Matplotlib, Scikit-image
 - Built a traditional CV model using pixel analysis, linear algebra, and geometric transforms to recognize simple object patterns.
 - Applied edge, Hough, contour, thresholding, and rule-based features to classify shapes without deep learning.
 - Used intermediate visualizations to debug feature extraction behavior and explain the recognition pipeline.

ACADEMIC AND PROJECT PRACTICES

- **Research-Oriented Development:** Breaks complex AI/ML systems into reproducible pipelines, baseline experiments, evaluation metrics, and iterative improvement cycles.
- **Model Evaluation:** Uses quantitative metrics such as PSNR, MSE, and SSIM for image-restoration tasks, along with qualitative visual inspection for artifact analysis.
- **System Design:** Focuses on modular components, clear data flow, scalable backend logic, secure execution boundaries, and practical user-facing workflows.
- **Documentation:** Prepares structured README files, project roadmaps, implementation prompts, architecture notes, and professor-facing academic reports.

RELEVANT COURSEWORK AND INTERESTS

- **Coursework:** Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Image Processing, Web Development.
- **Interests:** Computer Vision, Image Restoration, Multimodal AI, Local AI Agents, Human-Computer Interaction, Full-Stack AI Applications, Secure Automation Systems.